

MANISH HIRA

Data Science & Analytics Leader | AI/ML Engineering & Strategic Innovation

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Executive Summary

Senior Data Science and Analytics Leader with 15+ years of experience steering high-impact AI/ML engineering, predictive modeling, and enterprise data strategy across financial services (Capital One) and global technology firms (Samsung, Wipro). Ph.D. Candidate (VCU), M.S. in Computational Finance (UW), and MBA (Notre Dame). Proven track record of informing \$50M+ marketing allocations, authoring 3 granted patents, leading cross-functional teams, and driving enterprise GenAI and agentic workflow adoption.

Core Competencies & Technical Skills

- **Machine Learning & Statistics:** XGBoost, PyTorch, TensorFlow, Scikit-Learn, Sufficient Dimension Reduction (SDR), Causal Inference, Time Series Analysis (GARCH), A/B Testing, Feature Engineering, Bayesian Networks
- **Big Data & Cloud Architecture:** Snowflake, Databricks, AWS (Redshift, EMR, Glue, Lambda, SageMaker), PySpark, SQL, Python, R, C/C++, Unix Shell, ETL/ELT Pipelines
- **AI Innovation & GenAI Tooling:** Agentic Workflows (Claude Code, Gemini Gems), Copilot, LLaMA Sandbox Integration, Data Governance Automation, Model Audit & Validation
- **Leadership & Business Strategy:** Portfolio Management, Executive Communications, Agile/JIRA, Cross-Functional Team Leadership, Vendor Data Management, Product & Brand Intelligence

Professional Experience

Capital One, Richmond, VA

January 2019 – Present

Data Analysis Manager

- Led strategic analytical initiatives for brand and marketing intelligence, developing quantitative models and insights that directly informed executive decision-making.
- Evaluated multi-market advertising experiments across 10 major launch regions, directly optimizing \$50M in media allocation.
- Authored 20-year media investment analysis that established baseline brand marketing spend at 20% of total budget.
- Built predictive market opportunity model using XGBoost combined with Experian TAPS credit spending data.
- Spearheaded enterprise AI tooling adoption across team: deployed custom Gemini Gems for standardized tech templates and data governance, and integrated GitHub Copilot and Claude Code for agentic workflow optimization.
- Designed purchase volume (PVOL) geographic market targeting analysis, compressing delivery time from weeks to days for urgent executive decisions.
- Led data governance initiative migrating unstructured vendor data from spreadsheet silos to Snowflake, eliminating compliance risks and securing timeline execution.
- Resolved critical big-data bottlenecks on high-volume consumer clickstream datasets, optimizing architecture for downstream data science pipelines.
- Managed a team of analysts and data engineers, overseeing talent development, performance management, and Program Increment (PI) technology roadmaps.

State of Virginia / Capital One, Richmond, VA

November 2015 – December 2018

Data Analytics Consultant

- Executed data science and analytics consulting engagements for financial services and retail enterprise clients.
- Designed targeted card marketing campaigns featuring customer segmentation for 50M+ consumers and \$500M+ quarterly transaction volume.
- Developed XGBoost predictive models (AUC: 0.81) utilizing credit card transaction data to identify first-time retail partner shoppers.
- Engineered cloud-based ETL/ELT data pipelines from operational staging to enterprise data warehouse using Unix scripting and Oracle.

Samsung Electronics, South Korea & India

January 2009 – August 2013

Senior Manager

- Directed product portfolio planning for mobile division, directly impacting global sales of ~100M smartphone units.
- Founded Intellectual Property (IP) Cell for Innovation Management, originating an institutional IP pipeline that yielded 18 granted patents.
- Built enterprise Business Intelligence (BI) infrastructure and dashboards utilizing MS SQL Server and SSIS/SSRS.
- Led and mentored an engineering team developing embedded software for mobile devices.

- Managed \$10M+ multimedia software development for Qualcomm, resolving OEM commercialization blockers for MSM7x00 chipsets.
- Engineered automated mobile testing framework for Nokia Mobile (Finland) using Linux open-source software on TI OMAP platforms.
- Coordinated cross-border development of \$2M DICOM-compliant MRI scanning software for Toshiba.

Education

Virginia Commonwealth University, Richmond, VA
Ph.D. Candidate, Systems Modeling and Analysis

2024 – Present

- Joint Statistical Sciences, Operations Research & Applied Mathematics program. Research focus: Sufficient Dimension Reduction (SDR) and causal inference in high-dimensional financial modeling.

University of Washington, Seattle, WA

2019 – 2021

M.S., Computational Finance & Risk Management (GPA: 3.7)

- *Best Project Award*: Mortgage Default Credit Risk Model using ML (Fannie Mae dataset; XGBoost, Random Forest, DNN).

University of Notre Dame, Mendoza College of Business, Notre Dame, IN
MBA, Consulting & Finance (*Graduated Cum Laude*, GPA: 3.6 | GMAT: 750)

2013 – 2015

- *Arthur Foundation Fellow* (100% merit tuition scholarship).

University of Delhi, Netaji Subhas Institute of Technology (NSIT), Delhi, India
B.E., Instrumentation & Control Engineering

1996 – 2000

Granted Patents, Key Publications & Presentations

- **Granted US & International Patents:**
 - *Smart Card Authentication System*: US Patents 11,836,733 & 12,154,116 (Capital One Services, LLC).
 - *Near Field Communication (NFC) Method & System*: US Patent 9,160,418 & India Patent IN954DE2012 (Samsung Electronics).
- **Selected Publications & Keynotes:**
 - *Scientific Culture (2026)*: “Authenticity-Creativity Trade-offs in Generative AI Representations of Cultural Heritage: Voice AI Integration in Music Reconstruction” (Vol. 12, No. 5, pp. 2719–2733).
 - *MSI/ARF Analytics & Forecasting Conference (NYC, Sep 2026)*: Accepted Keynote Presentation — “GEO in the Loop: Integrating Generative Engine Optimization into Marketing Analytics Workflows.”
 - *Well Testing Journal (2025)*: “Volatility Modeling Using GARCH and Machine Learning Hybrids” (Vol. 34, Issue S3).
 - *Journal of Artificial Intelligence & Research Applications (2025)*: “Role of Machine Learning and AI in Real-Time Ad Personalization.”
 - *IEEE AISP Conference Proceedings (2026)*: “Integrating Deep Learning and Traditional Techniques for Image Enhancement.”

Executive Service, Industry Advisory & Honors

- **Executive & Industry Judging:** Judge, CODiE Awards (Best AI Data & Analytics Solution, 2025–2026) | Judge, AI Excellence Awards (Business Intelligence Group, 2026) | Judge, CES Awards (2025) | Judge, AI for Good Hackathon (2025).
- **Committee & Peer Review Leadership:** Committee Member, Capital One Modeling & Analytics Conference (MAC 2026) Breakouts Committee | Peer Reviewer & Technical Program Committee for International & IEEE Conferences (EDIGE, NILES, ICNCCom, ICIIBMS, ICCSTE, SPML, iGET, AMLDS, TEMSMET, ICEECE).
- **Professional Affiliations & Pro-Bono Advisory:** IEEE Senior Member | Association of National Advertisers (ANA) Committees | Pi Mu Epsilon Mathematics Honor Society | Pro-bono Data & Technology Consultant via Taproot Foundation & Startup Virginia.

Professional Certifications & Specializations

- **Cloud & Big Data Certifications:** AWS Certified Machine Learning – Specialty | AWS Certified Data Analytics – Specialty.

- **Advanced AI & Quantitative Certifications:** Explainable AI (XAI) Specialization (Duke University) | PyTorch for Deep Learning (DeepLearning.AI) | MLOps Specialization (Coursera) | Quantum Computing & Programming (QBronze) | Machine Learning Engineer Nanodegree (Udacity).